

- 10 a x -intercept = 1, y -intercept = 4
 b x -intercept = 9, y -intercept = 3
 c x -intercept = 10, y -intercept = -5
 d x -intercept = -3, y -intercept = -6
- 11 a $6x + 3y = 12$ b $2x - 6y = 2$
 c $-x + 8y = -6$ d $-16x + 20y = 0$

4A Solving linear equations

4A Start thinking!

- 1 a, c, f and g are linear equations
 2 a subtract 9 b $x = 7$
 3 a i add 4 ii divide by 3
 b You would get an incorrect solution if you did not perform the 'undo' operations in the reverse order to that used to form the equation.
 c $3x - 4 = 2$ d $3x - 4 = 2$
 $3x - 4 + 4 = 2 + 4$ $3x = 6$
 $3x = 6$ $x = 2$
 $\frac{3x}{3} = \frac{6}{3}$
 $x = 2$
 e Substitute the solution into the original equation to check left side equals right side.
 LS = $3 \times 2 - 4 = 2 =$ RS
- 4 a multiply by 5 then add 2; $x = -18$
 b subtract 7 then divide by -4; $x = -2$

Exercise 4A Solving linear equations

- 1 a $x = 5$ b $x = 1$ c $x = 15$ d $x = -5$
 e $x = 8$ f $x = 10$ g $x = 6$ h $x = -4$
 i $x = 9$ j $x = -7$ k $x = 3$ l $x = 2$
 m $x = 8$ n $x = -1$ o $x = 0$
- 2 a $x = 10$ b $x = -4$ c $x = 5$ d $x = -8$
 e $x = 7$ f $x = 6$ g $x = 8$ h $x = -2$
 i $x = 3$
- 3 a $x = -8\frac{1}{2}$ b $x = 1.7$ c $x = 4$
 d $x = -2.464$ e $x = 2.4$
 f $x = -9\frac{1}{2}$ or -9.5
- 4 a $x = -5$ b $x = 3$ c $x = -2$ d $x = 6$
 e $x = -\frac{1}{3}$ f $x = 7$ g $x = 3$ h $x = \frac{1}{9}$
 i $x = 6$
- 5 a $x = 2$ b $x = -5$ c $x = -6$ d $x = -4$
 e $x = 4$ f $x = 7.5$ g $x = 3\frac{1}{3}$ h $x = \frac{2}{35}$
 i $x = 2$ j $x = -1$ k $x = \frac{1}{4}$ l $x = -2\frac{2}{5}$
- 6 a $x = 2$ b $x = 5$ c $x = -3$ d $x = 1$
 e $x = -7$ f $x = -2$ g $x = 3$ h $x = 4$
 i $x = \frac{3}{7}$ j $x = -4$ k $x = 2$ l $x = -5$
- 7 a $x = 5$ b $x = -2$ c $x = 3$ d $x = -1$
 e $x = 6$ f $x = 2$ g $x = -4$ h $x = 8$
 i $x = 4$ j $x = -1$ k $x = 5$ l $x = 9$
- 8 a $x = -8$ b $x = 4$ c $x = -2$ d $x = 2$
 e $x = -1$ f $x = 4$
- 9 a $x = -8$ b $x = 31$ c $x = 8$ d $x = 1$
 e $x = 17$ f $x = 13$ g $x = -4\frac{2}{3}$ h $x = 6\frac{1}{7}$
 i $x = -2\frac{1}{8}$
- 10 a $x = b - a$ b $x = k + p$ c $x = \frac{d}{c}$
- d $x = \frac{h-g}{3}$ e $x = \frac{y-5}{4}$ f $x = mn - a$
 g $x = \frac{5+2w}{7}$ h $x = a(b-c)$ i $x = \frac{h}{k}$
 j $x = \frac{e+f}{2}$ k $x = \frac{v-2y}{2}$ l $x = p(w-q)$
 m $x = \frac{a}{b+c}$ n $x = \frac{2ny+m}{k}$ o $x = \frac{5c-3d}{8}$
- 11 a let n = number of weeks of saving
 b $24n + 105 = 219$ c $n = 4.75$
 d It will take Tom 5 weeks to save for his new tennis racquet.
- 12 a \$89.50 b length: 28 m, width: 16 m
 c Lisa: 14 goals, Nicole: 9 goals, Tania: 6 goals
 d 51 people
- 13 a $6d + 3.8 = 4d + 9.2$, where d is cost of a dumpling
 b \$2.70
- 14 a $A = 50 \text{ m}^2$ b $h = 7 \text{ mm}$ c $a = 7 \text{ m}$
 d $a = \frac{2A}{h} - b$
 e i $a = 52 \text{ cm}$ ii $a = 7.6 \text{ m}$
- 15 a $v = 34 \text{ m/s}$ b $t = 5.5 \text{ s}$
- 16 15, 17, 19, 21, 23
- 17 $n - 25 - \frac{n-25}{3} = 40$, where n is number of cupcakes made; 25 chocolate, 20 lemon, 40 without icing and 85 in total
- 18 a $x = -6$ b $x = 8$ c $x = -1$ d $x = 13$
 e $x = \frac{1}{8}$ f $x = -2\frac{7}{10}$

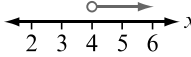
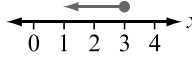
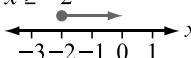
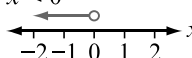
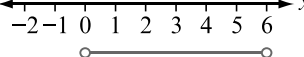
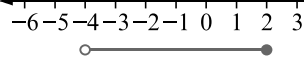
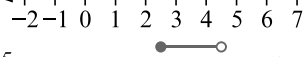
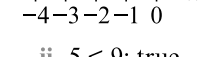
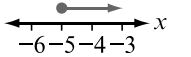
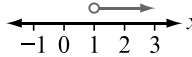
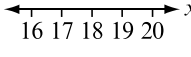
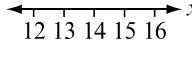
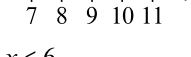
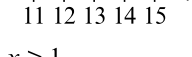
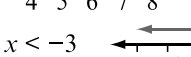
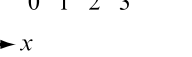
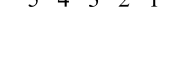
4B Solving linear inequalities

4B Start thinking!

- 1 < (less than), $\leq$ (less than or equal to),
 > (greater than), $\geq$ (greater than or equal to)
- 2 a > b < c < d >
- 3 a x is greater than 2
 b x is less than or equal to 3
 c x is greater than or equal to 5
 d x is less than -4
- 4 Each inequality represents a region on the number line and not a single point.
- 5 Some possible answers are given.
 a 7, 30.5, 122 b 4, 0, -5
 c -3, -1, 10.6 d -5, -10, -36.4
- 6 x is greater than 2, and less than 9. Some possible values are: 4, 5.7, 8.2.
- 7 -4, -3, -2, -1, 0, 1, 2

Exercise 4B Solving linear inequalities

- 1 a 2.4, 3, 7, 8.3, $6\frac{4}{5}$
 b -5, -1.2, 0, $-\frac{3}{4}$, -10, -4.9
 c 7, 8.3, $6\frac{4}{5}$ d -5, -10, -4.9
 e 2.4, 3, 7, $6\frac{4}{5}$ f $2.4, \frac{1}{2}, 3, 6\frac{4}{5}$
 g -5, 2.4, -1.2, $\frac{1}{2}$, 0, $-\frac{3}{4}$, -4.9
 h -5, -1.2, $-\frac{3}{4}$, -4.9
- 2 a $x \leq 5$ b $x > 1$ c $x \geq -4$
 d $x < -2$ e $3 \leq x \leq 7$ f $-5 < x \leq 0$

- 3 a $x > 4$  b $x \leq 3$ 
- c $x \geq -2$  d $x < 0$ 
- e $1 \leq x \leq 5$  f $-4 < x < 2$ 
- g $0 < x \leq 6$  h $-2.5 \leq x < -0.5$ 
- 4 a i $3 < 7$; true ii $5 < 9$; true
 iii $9 < 13$; true iv $-2 < 2$; true
 b i $1 < 5$; true ii $-1 < 3$; true
 iii $-5 < -1$; true iv $6 < 10$; true
 c i $4 < 12$; true ii $-4 < -12$; not true
 iii $6 < 18$; true iv $-2 < -6$; not true
 d i $1 < 3$; true ii $-1 < -3$; not true
 iii $\frac{2}{3} < 2$; true iv $-2 < -6$; not true
 e Inequality sign stays the same when adding or subtracting any number. Also stays the same when multiplying or dividing by a positive number. However, direction of inequality sign is reversed when multiplying or dividing by a negative number.
 f Some possible answers are given.
 i $4 > -8$, $4 + 2 > -8 + 2$ ($6 > -6$),
 $4 - 1 > -8 - 1$ ($3 > -9$)
 ii $4 > -8$, $4 \times 2 > -8 \times 2$ ($8 > -16$),
 $4 \div 2 > -8 \div 2$ ($2 > -4$)
 iii $4 > -8$, $4 \times -2 < -8 \times -2$ ($-8 < 16$),
 $4 \div -2 < -8 \div -2$ ($-2 < 16$)
- 5 a $x + 3 > 8$ b $-2x \geq 8$ c $x \leq 10$
 d $x < 4$ e $-x \geq -9$ f $x > -16$
 g $x \geq 7$ h $-x > 3$ i $-6x > -5$
- 6 a $x < 5$ b $x \geq 7$ c $x \leq -6$
 d $x > 6$ e $x \leq -5$ f $x < 50$
 g $x \geq -20$ h $x \leq 2$ i $x > -5$
- 7 a $x \geq -6$ b $x < -2$ c $x \leq 3$
 d $x > -24$ e $x < -7$ f $x \geq 1$
 g $x > -24$ h $x < 7$ i $x \leq 10$
- 8 a $x \geq -5$  b $x > 1$ 
- c $x \geq 18$  d $x < 14$ 
- e $x \leq 10$  f $x > 13$ 
- g $x \leq 6$  h $x > 1$ 
- i $x < -3$ 

- 9 a $x > 4$ b $x \leq -2$ c $x \geq 3$
 d $x > 4$ e $x < 1$ f $x \geq -1$
 g $x \leq 5$ h $x < 2$ i $x > \frac{1}{2}$
- 10 a $x > 11$ b $x \leq 13$ c $x \geq -2$
 d $x > -19$ e $x > -15$ f $x \leq 1$
 g $x \leq 3$ h $x > -\frac{1}{4}$ i $x \geq -6.5$
- 11 a $x \leq -7$ b $x > 1$ c $x \leq -3$
 d $x < 1$ e $x \geq 4$ f $x > -5$
- 12 a one
 b Linear inequalities can have many solutions within a given range.
- 13 a $p \geq 650\,000$, where p is selling price in \$
 b $h > 97$, where h is height of a person in cm
 c $s \leq 60$, where s is speed in km/h
 d $125 \leq h \leq 196$, where h is height of a person in cm
- 14 a $3p \leq 20$, where p is cost of pack of gum in \$;
 $p \leq 6\frac{2}{3}$
 b Todd could buy 1, 2, 3, 4, 5 or 6 packs of gum.
- 15 a $m = \frac{1}{2}(20 - x)$, where m is number of watermelons they will each take home.
 b $\frac{1}{2}(20 - x) \leq 3$; $x \geq 14$
 c 14 or more watermelons
- 16 a 25 min b 16 min c 19 min to 29 min
- 17 a $x \geq -11$ b $x < 4$ c $x \geq 2$ d $x < -8$
 e $x \leq 2$ f $x \leq -3$
- 18 a $1 \leq x \leq 6$ b $-4 > x > -7$

4C Sketching linear graphs

4C Start thinking!

- The graph of a linear relationship is a straight line.
 - A straight line connects two points, and this will define the linear relationship.
- 3 a i $m = 4$, $c = 1$ ii $m = -4$, $c = 1$
 iii $m = \frac{1}{4}$, $c = 1$
 b i (0, 1) ii (0, 1) iii (0, 1)
 c i 4 ii -4 iii $\frac{1}{4}$
 d i rise = 4, run = 1 ii rise = -4, run = 1
 iii rise = 1, run = 4
 e i $y = 4x + 1$ matches graph C
 ii $y = -4x + 1$ matches graph A
 iii $y = \frac{1}{4}x + 1$ matches graph B
- 4 Starting at y -intercept, use rise and run to locate a second point. Ruling a straight line passing through these two points will represent the graph of the linear relationship.

Exercise 4C Sketching linear graphs

